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LOW EARTH ORBIT (LEO) SATELLITE COMMUNICATIONS SYSTEM AND METHOD

Abstract

Aspects of the subject disclosure may include, for example, a method comprising: determining whether a backhaul communication path between a wireless communications node and a core network has become degraded; responsive to the backhaul communication path having become degraded, connecting a low earth orbit (LEO) satellite antenna for first bi-directional communication between the LEO satellite antenna and a radio element of the node; configuring the LEO satellite antenna for second bi-directional communication between the LEO satellite antenna and a LEO satellite; and based upon the configuring, facilitating outgoing communications from each of a plurality of wireless end-user devices to the core network via the LEO satellite and facilitating incoming communications to each of the plurality of wireless end-user devices from the core network via the LEO satellite. Other embodiments are disclosed.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] The subject disclosure relates to a Low Earth Orbit (LEO) satellite communications system and method.

BACKGROUND

[0002] Geostationary Equatorial Orbit (GEO) satellites have been a staple in emergency communications for decades. Historically, GEO satellites have operated in the Ku or Ka bands to provide emergency communications when terrestrial wired/wireless networks are impaired. While GEO systems have proven reliable, they are difficult, as well as expensive, to operate and maintain. In addition to the high maintenance and operational cost, GEO systems are traditionally limited in terms of bandwidth. Inefficiencies and difficulties associated with GEO satellite systems typically make it a less desirable solution for large scale operations.

[0003] A specific example of a type of Ku/Ka GEO system is the Very Small Aperture Terminals (VSAT) system, which has been used for emergency communications in areas where terrestrial communications systems have been impaired (e.g., natural or manmade disasters). Furthermore, GEO systems have been deployed in rural areas where other terrestrial backhaul methods may be cost prohibitive.

[0004] Moreover, the use of GEO satellites for cellular communications is well established. The network elements contained in a cell site are connected to a satellite network as opposed to fiber optic or microwave. Generally, there is no traffic exclusion over such a satellite connection (i.e., the end-user can expect the same service regardless of the backhaul). For example, voice, data, and/or video streaming are transported over the satellite network and onto a service provider core network.

[0005] This existing mode of operation has been successful for many years; however, as mentioned above, GEO-centric satellite communications networks are typically difficult and expensive to maintain and operate. Each GEO satellite network is typically comprised of three essential segments: the teleport, the remote terminal, and the satellite. The teleport is the aggregate point where all the remote terminals connect into. The remote terminal is the antenna and satellite equipment at the remote end of the satellite system. The remote location could be, for example, a cell site or a SatCOLT (Satellite Cell on Light Truck/Trailer). The satellite is what links the teleport and the remote terminal.

[0006] Further, as mentioned above, GEO satellite bandwidth required to provide service is typically limited and expensive. The large distances between earth and the GEO satellite introduce latency. Data-centric applications can be significantly impacted by latency, causing dropped packets and retransmissions further clogging the connection. Weather and nature also become a factor; heavy rain events and smoke can cause rain-fade, solar flares can cause outages. GEO satellite networks require high power transmitters and complex coding mechanisms for the originating or terminating signal to be useful. The equipment required is expensive to purchase and maintain. Additionally, placing a GEO satellite in orbit is very expensive. Such satellites typically take years to develop and require hundreds of thousands of man hours to go from concept to launch. All these reasons combined are contributing factors to the excessive cost of GEO satellite bandwidth.

[0007] As seen, the infrastructure that makes up both sides of a GEO satellite system is complex, expensive, and difficult to maintain. Teleports require large spaces for hub equipment and the larger

antennas, while the remote side also requires large space to accommodate the VSAT dish and the terminating equipment. The equipment on the hub side of the satellite system is typically a system comprised of servers, modulators, demodulators, and routers. The hub equipment is responsible to receive and transmit signals to and from the remote terminals as well as connect into the core network. The other end of the satellite network, the remote side, is also comprised of complex and expensive network elements to maintain and operate. The VSAT dishes at the remote terminals are typically not stationary, they travel mounted in trucks or are packaged in hard cases for shipping. Moving these heavy dishes around can cause some of the components to become damaged. Many of these components are often proprietary and are expensive and difficult to obtain—therefore, failures can cause significant delays during restoration activities.

[0008] A different type of system utilizes Low Earth Orbit (LEO) satellites. Technological advances in rocket launching capabilities have opened the door for LEO satellite service providers to launch the required number of satellites to essentially blanket the earth's surface. The densification of LEO satellites has generated significant coverage improvements in the United States (as well as across the world). Depending on use-case, LEO satellite systems can be more effective and less expensive to operate in contrast to GEO satellite systems. Generally, LEO systems allow for much higher throughputs while potentially reducing operational and maintenance costs.

[0009] Of note, LEO satellite technology works significantly different than GEO satellite systems. LEO satellite service is provided via a fleet of satellites orbiting earth between 300 and 2000 km above earth. Unlike GEO satellites, LEO satellites move around earth instead of remaining on a stationary (relative to the earth) orbit. The coverage area of each satellite is significantly reduced in comparison to GEO satellites. This is due to the satellite's proximity to earth; therefore, many satellites are required to provide blanket coverage. LEO satellites are typically more economical to build and deploy compared to GEO satellites resulting in increased operational cost savings for end-users. Economies of scale in satellite building and deployment technologies are accelerating LEO operators' ability to solve the world's high speed connectivity gaps in rural and remote parts of the world at a fraction of the cost.

[0010] Table 1 below provides a comparison between various aspects of a typical LEO system and a typical GEO system.

TABLE-US-00001	TABLE 1	LEO system	GEO System
Dish Size	1.6 ft.sup.2-3.16 ft.sup.2	48 ft.sup.2-190 ft.sup.2	Weight
6.4-13 Lbs.	135-250 Lbs.	Antenna Cost	\$500-\$2500
\$65,000-\$180,000	Electronics Cost	\$25,000	\$35,000
Latency	<100 milli-seconds	>600 milli-seconds	Throughput DL
150 Mbps-300 Mbps*	Up to 40 Mbps*	Throughput UL	20 Mbps-30 Mbps*
Up to 20 Mbps*	Physical Connections	<10	23
*Subject to change and dependent on network conditions. Not guaranteed rate.			

[0011] In conclusion, GEO satellite systems have performed extremely well over the past decades but have fallen out of favor in recent years due largely to advances in LEO technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0013] FIG. 1 is a block diagram illustrating an example, non-limiting embodiment of a communication network in accordance with various aspects described herein.

[0014] FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a system (which can function fully or partially within the communication network of FIG. 1) in accordance with various aspects described herein.

[0015] FIG. 2B is a block diagram illustrating an example, non-limiting embodiment of a system (which can function fully or partially within the communication network of FIG. 1) in accordance with various aspects described herein.

[0016] FIG. 2C is a block diagram illustrating an example, non-limiting embodiment of a system (which can function fully or partially within the communication network of FIG. 1) in accordance with various aspects described herein.

[0017] FIG. 2D depicts an illustrative embodiment of a method in accordance with various aspects described herein.

[0018] FIG. 2E depicts an illustrative embodiment of a method in accordance with various aspects described herein.

[0019] FIG. 2F depicts an illustrative embodiment of a method in accordance with various aspects described herein.

[0020] FIG. 3 is a block diagram illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein.

[0021] FIG. 4 is a block diagram of an example, non-limiting embodiment of a computing environment in accordance with various aspects described herein.

[0022] FIG. 5 is a block diagram of an example, non-limiting embodiment of a mobile network platform in accordance with various aspects described herein.

[0023] FIG. 6 is a block diagram of an example, non-limiting embodiment of a communication device in accordance with various aspects described herein.

DETAILED DESCRIPTION

[0024] The subject disclosure describes, among other things, illustrative embodiments for Low Earth Orbit (LEO) satellite communications systems and methods. In various examples, the LEO satellite communications systems and methods can be for emergency use and/or can be in the form of a universal system and method. In one example, the LEO satellite emergency communications system and method can facilitate FirstNet usage. In one example, the LEO satellite emergency communications system and method can facilitate consumer cellular backhaul. Other embodiments are described in the subject disclosure.

[0025] One or more aspects of the subject disclosure include a method, comprising: determining whether a backhaul communication path between a wireless communications node and a core network has become degraded, wherein the wireless communications node comprises at least one radio element configured for wireless communications with a plurality of wireless end-user devices, and wherein the determining results in a first determination; responsive to the first determination being that the backhaul communication path has become degraded, connecting a Low Earth Orbit (LEO) satellite antenna for first bi-directional communication between the LEO satellite antenna and the at least one radio element; configuring the LEO satellite antenna for second bi-directional communication between the LEO satellite antenna and a LEO satellite, wherein the second bi-directional communication comprises a first uplink to the LEO satellite and a first downlink from the LEO satellite, wherein the LEO satellite is configured for third bi-directional communication with the core network, and wherein the third bi-directional communication comprises a second downlink from the LEO satellite and a second uplink to the LEO satellite; and based upon the configuring, facilitating outgoing communications from each of the plurality of wireless end-user devices to the core network via the LEO satellite and facilitating incoming communications to each of the plurality of wireless end-user devices from the core network via the LEO satellite.

[0026] One or more aspects of the subject disclosure include a method, comprising: determining whether a backhaul communication path between a cell site and a core service provider network has become degraded, wherein the cell site comprises at least one cellular radio configured for wireless communications with at least one end-user mobile communication device, wherein the cell site further comprises at least one baseband unit and at least one router, and wherein the

determining results in a first determination; responsive to the first determination being that the backhaul communication path has become degraded, installing a Low Earth Orbit (LEO) satellite antenna at the cell site for first bi-directional communication between the LEO satellite antenna and the at least one cellular radio via the at least one router and the at least one baseband unit; configuring the LEO satellite antenna for second bi-directional communication between the LEO satellite antenna and a LEO satellite, wherein the second bi-directional communication comprises a first uplink to the LEO satellite and a first downlink from the LEO satellite, wherein the LEO satellite is configured for third bi-directional communication with the core service provider network, and wherein the third bi-directional communication comprises a second downlink from the LEO satellite and a second uplink to the LEO satellite; and based upon the configuring, facilitating an outgoing communication from the at least one end-user mobile communication device to the core service provider network via the LEO satellite and facilitating an incoming communication to the at least one end-user mobile communication device from the core service provider network via the LEO satellite.

[0027] One or more aspects of the subject disclosure include a method, comprising: responsive to determining that a backhaul communication path between a cellular base station and a core network has become degraded, installing a Low Earth Orbit (LEO) satellite antenna on a tower of the cellular base station for first bi-directional communication between the LEO satellite antenna and a cellular radio of the cellular base station, wherein the first bi-directional communication is via a router of the cellular base station and a baseband unit of the cellular base station, and wherein the cellular radio is configured for wireless communications with a plurality of end-user cellular communication devices; configuring the LEO satellite antenna for second bi-directional communication between the LEO satellite antenna and a LEO satellite, wherein the second bi-directional communication comprises a first uplink to the LEO satellite and a first downlink from the LEO satellite, wherein the LEO satellite is configured for third bi-directional communication with the core network, and wherein the third bi-directional communication comprises a second downlink from the LEO satellite and a second uplink to the LEO satellite; and based upon the configuring, implementing outgoing communications from the plurality of end-user cellular communication devices to the core network via the LEO satellite and implementing incoming communications to the plurality of end-user cellular communication devices from the core network via the LEO satellite.

[0028] Referring now to FIG. 1, a block diagram is shown illustrating an example, non-limiting embodiment of a system **100** in accordance with various aspects described herein. For example, system **100** can facilitate in whole or in part a LEO satellite emergency communications system and method. In particular, a communications network **125** is presented for providing broadband access **110** to a plurality of data terminals **114** via access terminal **112**, wireless access **120** to a plurality of mobile devices **124** and vehicle **126** via base station or access point **122**, voice access **130** to a plurality of telephony devices **134**, via switching device **132** and/or media access **140** to a plurality of audio/video display devices **144** via media terminal **142**. In addition, communication network **125** is coupled to one or more content sources **175** of audio, video, graphics, text and/or other media. While broadband access **110**, wireless access **120**, voice access **130** and media access **140** are shown separately, one or more of these forms of access can be combined to provide multiple access services to a single client device (e.g., mobile devices **124** can receive media content via media terminal **142**, data terminal **114** can be provided voice access via switching device **132**, and so on).

[0029] The communications network **125** includes a plurality of network elements (NE) **150**, **152**, **154**, **156**, etc. for facilitating the broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or the distribution of content from content sources **175**. The communications network **125** can include a circuit switched or packet switched network, a voice over Internet protocol (VOIP) network, Internet protocol (IP) network, a cable network, a passive or active

optical network, a 4G, 5G, or higher generation wireless access network, WIMAX network, UltraWideband network, personal area network or other wireless access network, a broadcast satellite network and/or other communications network.

[0030] In various embodiments, the access terminal **112** can include a digital subscriber line access multiplexer (DSLAM), cable modem termination system (CMTS), optical line terminal (OLT) and/or other access terminal. The data terminals **114** can include personal computers, laptop computers, netbook computers, tablets or other computing devices along with digital subscriber line (DSL) modems, data over coax service interface specification (DOCSIS) modems or other cable modems, a wireless modem such as a 4G, 5G, or higher generation modem, an optical modem and/or other access devices.

[0031] In various embodiments, the base station or access point **122** can include a 4G, 5G, or higher generation base station, an access point that operates via an 802.11 standard such as 802.11n, 802.11ac or other wireless access terminal. The mobile devices **124** can include mobile phones, e-readers, tablets, phablets, wireless modems, and/or other mobile computing devices.

[0032] In various embodiments, the switching device **132** can include a private branch exchange or central office switch, a media services gateway, VoIP gateway or other gateway device and/or other switching device. The telephony devices **134** can include traditional telephones (with or without a terminal adapter), VOIP telephones and/or other telephony devices.

[0033] In various embodiments, the media terminal **142** can include a cable head-end or other TV head-end, a satellite receiver, gateway or other media terminal **142**. The display devices **144** can include televisions with or without a set top box, personal computers and/or other display devices.

[0034] In various embodiments, the content sources **175** include broadcast television and radio sources, video on demand platforms and streaming video and audio services platforms, one or more content data networks, data servers, web servers and other content servers, and/or other sources of media.

[0035] In various embodiments, the communications network **125** can include wired, optical and/or wireless links and the network elements **150**, **152**, **154**, **156**, etc. can include service switching points, signal transfer points, service control points, network gateways, media distribution hubs, servers, firewalls, routers, edge devices, switches and other network nodes for routing and controlling communications traffic over wired, optical and wireless links as part of the Internet and other public networks as well as one or more private networks, for managing subscriber access, for billing and network management and for supporting other network functions.

[0036] Referring now to FIG. 2A, this is a block diagram illustrating an example, non-limiting embodiment of a system **200** (which can function fully or partially within the communication network of FIG. 1) in accordance with various aspects described herein. As seen in this figure (which relates to a LEO emergency communications portable cell site backhaul connection diagram), an existing cell site **202** can include: one or more cellular radios **202A** which are in operative communication with one or more L3 baseband units **202B** which are in operative communication with one or more routers **202C** (the cell site can also include a tower (not identified with a call out number)). In normal operation, the cell site **202** facilitates bi-directional communication between a plurality of end-user mobile devices (not shown) and the core network **204** (of a cellular provider) via a backhaul communication path **206** (e.g., fiber optic communication path, microwave communication path). However, when the backhaul communication path **206** becomes degraded (e.g., unusable, broken, and/or non-existent) as shown in this figure by the “X” marking, then a LEO satellite can be installed (according to various embodiments) to provide supplemental (e.g., emergency) backhaul as described herein. In this regard, a LEO antenna **208** can be attached (e.g., physically mounted) to the tower of the cell site **202** and can be wired to receive power from the cell site **202**. Further, the LEO antenna **208** can be configured to facilitate bi-directional communications between the end-user mobile devices and the core network **204** via a path comprising the following operatively connected elements: cellular

radio(s) **202A**, baseband unit(s) **202B**, router(s) **202C**, LEO antenna **208**, LEO satellite **210**, LEO provider ground station **212**, LEO provider network **214**, Internet **216**, secure gateway **218**, and core network **204**. As shown in this figure, an uplink path **220A** and a downlink path **220B** are facilitated between LEO antenna **208** and LEO satellite **210**. Further, an uplink path **222A** and a downlink path **222B** are facilitated between LEO satellite **210** and LEO provider ground station **212**. Of course, various embodiments can operate in the context of a plurality of LEO satellites, wherein respective uplink/downlink paths are facilitated between each of a succession of LEO satellites (e.g., as each LEO satellite comes into view).

[0037] As described herein, various embodiments can utilize LEO CSBH (cell site backhaul) inter/intra connections as follows: (I) Satellite transport interconnection: (a) LEO Antenna Panel connects to power supply and router; (b) Router provides interconnection between Baseband Unit (BBU) and LEO satellite antenna; (II) Cellular equipment interconnection: (a) BBU connects to the router; (b) Individual radios connect to the BBU; (c) Cellular antennas connect into radios; (III) Cell site connection to core network: The radio equipment can connect (for example) to the core network through the open internet (via the LEO satellite system) through a service provider co-located POP.

[0038] As described herein, various embodiments can be deployed as follows: (a) A team of engineers deploys a LEO CSBH kit to a cell site; (b) The LEO antenna is mounted on the highest reachable point in the cell site where there is a clear view of the sky; (c) The preconfigured baseband unit is placed inside the shelter; (d) The existing cell site radios are connected to the preconfigured baseband unit; (e) Power is provided to each network component; (f) The radios are enabled; and (g) Call testing is conducted to verify that service is restored.

[0039] As described herein, various embodiments facilitate deployment of a LEO CSBH kit at a cell site where traditional transport is unavailable for one or more of the following reasons: (a) Natural/Manmade disaster; (b) Localized extended outage; and/or (c) Cost of fiber to the site is excessive.

[0040] As described herein, various embodiments can facilitate deployment simplicity, reduced size/weight of each kit, and/or reduced cost of each kit (e.g., to generate a repeatable process across multiple cell sites with minimal personnel).

[0041] Referring now to FIG. 2B, this is a block diagram illustrating an example, non-limiting embodiment of a system **230** (which can function fully or partially within the communication network of FIG. 1) in accordance with various aspects described herein. As seen in this figure (which relates to a LEO emergency communications portable cell site backhaul connection diagram), an existing cell site **232** can include: one or more cellular radios **232A** which are in operative communication with one or more L3 baseband units **232B** which are in operative communication with one or more routers **232C** (the cell site can also include a tower (not identified with a call out number)). In normal operation, the cell site **232** facilitates bi-directional communication between a plurality of end-user mobile devices (not shown) and the core network **234** (of a cellular provider) via a backhaul communication path **236** (e.g., fiber optic communication path, microwave communication path). However, when the backhaul communication path **236** becomes degraded (e.g., unusable, broken, and/or non-existent) as shown in this figure by the “X” marking, then a LEO satellite can be installed (according to various embodiments) to provide supplemental (e.g., emergency) backhaul as described herein. In this regard, a LEO antenna **238** can be attached (e.g., physically mounted) to the tower of the cell site **232** and can be wired to receive power from the cell site **232**. Further, the LEO antenna **238** can be configured to facilitate bi-directional communications between the end-user mobile devices and the core network **234** via a path comprising the following operatively connected elements: cellular radio(s) **232A**, baseband unit(s) **232B**, router(s) **232C**, LEO antenna **238**, LEO satellite **240**, LEO provider ground station **242**, LEO provider network **244**, Internet **246**, secure gateway **248**, and core network **234**. As shown in this figure, an uplink path **250A** and a downlink path **250B** are

facilitated between LEO antenna **238** and LEO satellite **240**. Further, an uplink path **252A** and a downlink path **252B** are facilitated between LEO satellite **240** and LEO provider ground station **242**. Of course, various embodiments can operate in the context of a plurality of LEO satellites, wherein respective uplink/downlink paths are facilitated between each of a succession of LEO satellites (e.g., as each LEO satellite comes into view). As seen, this system **230** is similar to system **200** of FIG. 2A, with the main difference being that in this system **230** the core network **234** can communicate with the LEO provider network **244** (via the secure gateway **248**) using a secure tunnel, a VPN connection, or the like (see arrow “A” connecting secure gateway **248** to LEO provider network **244**).

[0042] Referring now to FIG. 2C, this is a block diagram illustrating an example, non-limiting embodiment of a system **270** (which can function fully or partially within the communication network of FIG. 1) in accordance with various aspects described herein. As seen in this figure (which relates to LECP—Integrated Cased Solution—Connection Diagram), a portable modular kit **272** can include: one or more small cell radios **272A** which are in operative communication with one or more L3 baseband units **272B** which are in operative communication with one or more routers **272C** which are in operative communication with one or more LEO antennas **272D**. The portable modular kit **272** can facilitate bi-directional communication between a plurality of end-user mobile devices (not shown) and the core network **274** (of a cellular provider) where no such communication would otherwise be available. In this regard, the LEO antenna(s) **272D** can be configured to facilitate bi-directional communications between the end-user mobile devices and the core network **274** via a path comprising the following operatively connected elements: small cell radio(s) **272A**, baseband unit(s) **272B**, router(s) **272C**, LEO antennas(s) **272D**, LEO satellite **290**, LEO provider ground station **282**, LEO provider network **284**, Internet **286**, secure gateway **288** and core network **274**. As shown in this figure, an uplink path **292A** and a downlink path **292B** are facilitated between LEO antenna(s) **272D** and LEO satellite **290**. Further, an uplink path **294A** and a downlink path **294B** are facilitated between LEO satellite **290** and LEO provider ground station **282**. Of course, various embodiments can operate in the context of a plurality of LEO satellites, wherein respective uplink/downlink paths are facilitated between each of a succession of LEO satellites (e.g., as each LEO satellite comes into view).

[0043] As described herein, various embodiments can utilize LEO cased integrated cellular system interconnections as follows: (I) Satellite transport interconnection: (a) LEO Antenna Panel connects to power supply and router; (b) Router provides interconnection between Baseband Unit BBU and LEO satellite antenna; (II) Cellular equipment-BBU Transport interconnection: (a) Baseband Unit (BBU) connects to the router; (b) Individual radios connect to the BBU; (III) Cellular equipment-Small cellular radio connection to BBU case: (a) Small cell radios connect to BBU case via fiber; (b) Omnidirectional antennas connect to each small cell radio.

[0044] As described herein, in various embodiments a single portable modular kit (or cased unit) can implement both cellular backhaul at an existing cell site (see, e.g., FIGS. 2A and 2B as well as operate as a fully self-contained cell site (see, e.g., FIG. 2C).

[0045] Referring now to FIG. 2D, various steps of a method **2000** according to an embodiment are shown. As seen in this FIG. 2D, step **2002** comprises determining whether a backhaul communication path between a wireless communications node and a core network has become degraded, wherein the wireless communications node comprises at least one radio element configured for wireless communications with a plurality of wireless end-user devices, and wherein the determining results in a first determination. Next, step **2004** comprises responsive to the first determination being that the backhaul communication path has become degraded, connecting a Low Earth Orbit (LEO) satellite antenna for first bi-directional communication between the LEO satellite antenna and the at least one radio element. Next, step **2006** comprises configuring the LEO satellite antenna for second bi-directional communication between the LEO satellite antenna and a LEO satellite, wherein the second bi-directional communication comprises a first uplink to the

LEO satellite and a first downlink from the LEO satellite, wherein the LEO satellite is configured for third bi-directional communication with the core network, and wherein the third bi-directional communication comprises a second downlink from the LEO satellite and a second uplink to the LEO satellite. Next, step **2008** comprises based upon the configuring, facilitating outgoing communications from each of the plurality of wireless end-user devices to the core network via the LEO satellite and facilitating incoming communications to each of the plurality of wireless end-user devices from the core network via the LEO satellite.

[0046] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. 2D, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

[0047] Referring now to FIG. 2E, various steps of a method **2100** according to an embodiment are shown. As seen in this FIG. 2E, step **2102** comprises determining whether a backhaul communication path between a cell site and a core service provider network has become degraded, wherein the cell site comprises at least one cellular radio configured for wireless communications with at least one end-user mobile communication device, wherein the cell site further comprises at least one baseband unit and at least one router, and wherein the determining results in a first determination. Next, step **2104** comprises responsive to the first determination being that the backhaul communication path has become degraded, installing a Low Earth Orbit (LEO) satellite antenna at the cell site for first bi-directional communication between the LEO satellite antenna and the at least one cellular radio via the at least one router and the at least one baseband unit. Next, step **2106** comprises configuring the LEO satellite antenna for second bi-directional communication between the LEO satellite antenna and a LEO satellite, wherein the second bi-directional communication comprises a first uplink to the LEO satellite and a first downlink from the LEO satellite, wherein the LEO satellite is configured for third bi-directional communication with the core service provider network, and wherein the third bi-directional communication comprises a second downlink from the LEO satellite and a second uplink to the LEO satellite. Next, step **2108** comprises based upon the configuring, facilitating an outgoing communication from the at least one end-user mobile communication device to the core service provider network via the LEO satellite and facilitating an incoming communication to the at least one end-user mobile communication device from the core service provider network via the LEO satellite.

[0048] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. 2E, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

[0049] Referring now to FIG. 2F, various steps of a method **2200** according to an embodiment are shown. As seen in this FIG. 2F, step **2202** comprises responsive to determining that a backhaul communication path between a cellular base station and a core network has become degraded, installing a Low Earth Orbit (LEO) satellite antenna on a tower of the cellular base station for first bi-directional communication between the LEO satellite antenna and a cellular radio of the cellular base station, wherein the first bi-directional communication is via a router of the cellular base station and a baseband unit of the cellular base station, and wherein the cellular radio is configured for wireless communications with a plurality of end-user cellular communication devices. Next, step **2204** comprises configuring the LEO satellite antenna for second bi-directional communication between the LEO satellite antenna and a LEO satellite, wherein the second bi-directional communication comprises a first uplink to the LEO satellite and a first downlink from the LEO satellite, wherein the LEO satellite is configured for third bi-directional communication with the core network, and wherein the third bi-directional communication comprises a second

downlink from the LEO satellite and a second uplink to the LEO satellite. Next, step 2206 comprises based upon the configuring, implementing outgoing communications from the plurality of end-user cellular communication devices to the core network via the LEO satellite and implementing incoming communications to the plurality of end-user cellular communication devices from the core network via the LEO satellite. In another embodiment, the LEO satellite antenna can be installed at another location (instead of on the tower). For example, the LEO satellite antenna can be installed adjacent to the tower and/or on the top of a shelter at the base station.

[0050] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. 2F, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

[0051] As described herein, various embodiments of a LEO emergency cellular backhaul (CSBH) system provide a number of advantages. Such advantages include (but are not limited to): (a) Increased opportunities to reduce cellular downtime by restoring/maintaining cellular services to customers and first responders during and after a disaster; (b) Deployment simplicity (e.g., 90% lighter than GEO Ku Dish; 50% reduction of physical connections at deployment time (e.g., <10 in LEO system compared to >20 in GEO system); minimized failure points that increase deployment success rate; deployment time efficiency (e.g., <75 minutes); and/or service improvement (e.g., reduced latency by >75% on average; increased UL/DL data throughputs; increased number of cell site recoverability in area; increased cellular carrier support (e.g., two carriers with three sectors each)).

[0052] As described herein, various embodiments provide a solution that can be easily replicated as a repeatable process in most (if not all) areas across the country (and/or other countries) where there is an open sky. In one example, many of these kits can be staged across the United States at a fraction of the cost (due to the low costs of the components). Further, in addition to component and operational cost savings, units (according to various embodiments) can be designed to be easily deployed by most technicians. Mil-Spec style connectors can be used to limit the number of physical connections a field technician needs to make during deployment. This approach reduces failure points, and limits opportunities for a field team to make incorrect connections. Reducing the number of connections inherently increases deployment speed. For comparison, a GEO system typically requires twenty-three physical connections, whereas an LEO-ECP system (according to an embodiment) requires less than ten. Deployment simplicity, reduced equipment cost, and increased backhaul capabilities (according to various embodiments) make this an attractive solution.

[0053] As described herein, various embodiments provide a universal LEO satellite emergency communications system and method including FirstNet and consumer cellular backhaul.

[0054] As described herein, various embodiments use LEO satellite technology to provide cellular backhaul. Such embodiments using LEO systems are highly portable, inexpensive, and simple to set up. In various embodiments, cellular service can be quickly restored in areas impacted by disaster (and/or in remote areas where other backhaul options are not available). In various embodiments, a service provider is able to provide significantly better service due to the high speed and low latency of LEO satellite systems, at a fraction of the cost of GEO systems.

[0055] As described herein, various embodiments provide for connectivity recovery in the aftermath of a disaster.

[0056] As described herein, various embodiments can support cellular backhaul at one or more cell sites.

[0057] As described herein, various embodiments provide a “flyaway kit” that utilizes a LEO satellite system along with preconfigured equipment to quickly deploy at a cell site (and/or to bring

communications to remote areas where cellular coverage is otherwise non-existent). In various embodiments, these units can be deployed in large scale (because of reduced size and ease of deployment). Additionally, because of the low latency and high throughputs found in LEO satellite systems, various embodiments can provide a much-improved experience for end-users.

[0058] As described herein, various embodiments utilize LEO satellite technology for cell site backhaul and/or for portable cellular kits.

[0059] As described herein, various embodiments provide cellular backhaul in remote areas (e.g., where standard backhaul services are not available).

[0060] As described herein, due to advances in LEO services, various embodiments can benefit from the added efficiencies provided by LEO satellite technology.

[0061] As described herein, various embodiments can utilize LEO satellite technology as backhaul for macro cell sites. In one embodiment, using a radio baseband unit capable of communicating to a core network (e.g., over the open internet via a satellite connection), a cell site can be configured with backhaul.

[0062] As described herein, various embodiments can utilize the following components for a LEO backhaul solution: a satellite antenna/panel, a router, a baseband unit, and cellular radio(s). The satellite antenna/panel can include the antenna that connects to the LEO satellites. The router can allow the baseband unit to communicate with the satellite network. The baseband unit can manage the radio(s) connected to it. The radio(s) can broadcast the wireless cellular signal that allows the end-users to communicate to the cellular network.

[0063] As described herein, various embodiments can provide for a LEO Emergency Communications Portable (L-ECP) system. Such a L-ECP system can be a portable multi-purpose system that can be quickly deployed to a cell site to be used for cell site backhaul (and/or can be used as a miniature portable cell site using small cell technology).

[0064] As described herein, various embodiments can provide a L-ECP system for cell site backhaul. In one embodiment, existing radio(s) at a cell site can be connected to a preconfigured baseband unit to restore and/or establish cellular service.

[0065] As described herein, various embodiments can provide service (e.g., cellular service) in the aftermath of a disaster (e.g., a hurricane, an earthquake) when backhaul capabilities (e.g., fiber optic cable, microwave) of a terrestrial communications system become impaired.

[0066] As described herein, various embodiments can help restore communication services (e.g., cellular services) as quickly and as efficiently as possible after a natural or manmade disaster. The restored communication services can be used, for example, to request emergency services, communicate with family, file insurance claims, dispatch repair/restoration crews, coordinate first responders, and/or facilitate search, rescue, and recovery operations.

[0067] As described herein, various embodiments can provide a LEO cell site backhaul kit that can be quickly deployed (e.g., in under 30 minutes) to restore backhaul service in a given area. For instance, in a scenario where the tower infrastructure (e.g., tower, RF path, radios) are intact, a LEO cell site backhaul kit can be deployed to restore backhaul service in that area.

[0068] As described herein, in various embodiments a low latency, high-speed LEO satellite connection is provided to a core network.

[0069] As described herein, in various embodiments a field support team can install the LEO satellite antenna kit. The field support team can determine whether a LEO or a GEO is deployed (e.g., based on natural landscape such as tree coverage, mountains, etc.). The LEO solution can be the primary solution, while GEO can be a secondary solution.

[0070] As described herein, various embodiments can facilitate isolated regional service. For example, an ECP-LEO modular kit can be deployed to a remote isolated region where cellular coverage is otherwise non-existent. In one specific scenario, a small fire command basecamp can be supported in a remote area where there is otherwise no cellular coverage and there are no access roads to bring a conventional SatCOLT. In one embodiment, the kit can comprise the following

main parts: a satellite antenna, a transport/BBU case, two small cell radios.

[0071] As described herein, various embodiments can provide satellite backhaul to a cell site with an otherwise degraded (or non-functional) backhaul. In one embodiment, the field team: (a) Mounts the antenna on an elevated part of the cell site; (b) Connects the antenna case to the BBU case using a single cable; and (c) Connects each fiber to the existing cell site radios.

[0072] As described herein, various embodiments can provide a portable cell site for use in a remote setup. In one embodiment, the field team: (a) Mounts the antenna on an elevated part of the cell site; (b) Connects the antenna case to the BBU case using a single mil-spec terminated cable; and (c) Connects a single mil-spec terminated cable to the radio case.

[0073] As described herein, various embodiments can operate in a scenario where an existing cellular tower and an existing cellular infrastructure (e.g., macro cell site) are operational, but the transport is not operational (wherein a LEO system is then utilized to bring the cellular signal back into the core network).

[0074] As described herein, various embodiments can operate in a scenario where mechanisms are utilized to setup essentially an independent portable cell site that uses LEO for its backhaul.

[0075] As described herein, various embodiments can provide an emergency supplement backhaul capability.

[0076] As described herein, various embodiments can provide a system that's only used to provide transport and/or can provide a system with everything that is needed (e.g., including the radios and the transport).

[0077] As described herein, various embodiments can provide a modular kit comprising a LEO antenna, a radio baseband unit, and small cell radios to communicate with a LEO satellite.

[0078] As described herein, various embodiments can provide cellular radios that are connected (directly or indirectly) with a LEO antenna via one or more fiber optic cables.

[0079] Referring now to FIG. 3, a block diagram **300** is shown illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein. In particular a virtualized communication network is presented that can be used to implement some or all of the subsystems and functions of system **100**, some or all of the subsystems and functions of system **200**, some or all of the subsystems and functions of system **250**, and/or some or all of the functions of methods **2000**, **2100**, **2200**. For example, virtualized communication network **300** can facilitate in whole or in part a LEO satellite emergency communications system and method.

[0080] In particular, a cloud networking architecture is shown that leverages cloud technologies and supports rapid innovation and scalability via a transport layer **350**, a virtualized network function cloud **325** and/or one or more cloud computing environments **375**. In various embodiments, this cloud networking architecture is an open architecture that leverages application programming interfaces (APIs); reduces complexity from services and operations; supports more nimble business models; and rapidly and seamlessly scales to meet evolving customer requirements including traffic growth, diversity of traffic types, and diversity of performance and reliability expectations.

[0081] In contrast to traditional network elements-which are typically integrated to perform a single function, the virtualized communication network employs virtual network elements (VNEs) **330**, **332**, **334**, etc. that perform some or all of the functions of network elements **150**, **152**, **154**, **156**, etc. For example, the network architecture can provide a substrate of networking capability, often called Network Function Virtualization Infrastructure (NFVI) or simply infrastructure that is capable of being directed with software and Software Defined Networking (SDN) protocols to perform a broad variety of network functions and services. This infrastructure can include several types of substrates. The most typical type of substrate being servers that support Network Function Virtualization (NFV), followed by packet forwarding capabilities based on generic computing resources, with specialized network technologies brought to bear when general-purpose processors

or general-purpose integrated circuit devices offered by merchants (referred to herein as merchant silicon) are not appropriate. In this case, communication services can be implemented as cloud-centric workloads.

[0082] As an example, a traditional network element **150** (shown in FIG. **1**), such as an edge router can be implemented via a VNE **330** composed of NFV software modules, merchant silicon, and associated controllers. The software can be written so that increasing workload consumes incremental resources from a common resource pool, and moreover so that it is elastic: so, the resources are only consumed when needed. In a similar fashion, other network elements such as other routers, switches, edge caches, and middle boxes are instantiated from the common resource pool. Such sharing of infrastructure across a broad set of uses makes planning and growing infrastructure easier to manage.

[0083] In an embodiment, the transport layer **350** includes fiber, cable, wired and/or wireless transport elements, network elements and interfaces to provide broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or access to content sources **175** for distribution of content to any or all of the access technologies. In particular, in some cases a network element needs to be positioned at a specific place, and this allows for less sharing of common infrastructure. Other times, the network elements have specific physical layer adapters that cannot be abstracted or virtualized and might require special DSP code and analog front ends (AFEs) that do not lend themselves to implementation as VNEs **330**, **332** or **334**. These network elements can be included in transport layer **350**.

[0084] The virtualized network function cloud **325** interfaces with the transport layer **350** to provide the VNEs **330**, **332**, **334**, etc. to provide specific NFVs. In particular, the virtualized network function cloud **325** leverages cloud operations, applications, and architectures to support networking workloads. The virtualized network elements **330**, **332** and **334** can employ network function software that provides either a one-for-one mapping of traditional network element function or alternately some combination of network functions designed for cloud computing. For example, VNEs **330**, **332** and **334** can include route reflectors, domain name system (DNS) servers, and dynamic host configuration protocol (DHCP) servers, system architecture evolution (SAE) and/or mobility management entity (MME) gateways, broadband network gateways, IP edge routers for IP-VPN, Ethernet and other services, load balancers, distributors and other network elements. Because these elements do not typically need to forward large amounts of traffic, their workload can be distributed across a number of servers—each of which adds a portion of the capability, and which creates an elastic function with higher availability overall than its former monolithic version. These virtual network elements **330**, **332**, **334**, etc. can be instantiated and managed using an orchestration approach similar to those used in cloud compute services.

[0085] The cloud computing environments **375** can interface with the virtualized network function cloud **325** via APIs that expose functional capabilities of the VNEs **330**, **332**, **334**, etc. to provide the flexible and expanded capabilities to the virtualized network function cloud **325**. In particular, network workloads may have applications distributed across the virtualized network function cloud **325** and cloud computing environment **375** and in the commercial cloud or might simply orchestrate workloads supported entirely in NFV infrastructure from these third-party locations.

[0086] Turning now to FIG. **4**, there is illustrated a block diagram of a computing environment in accordance with various aspects described herein. In order to provide additional context for various embodiments of the embodiments described herein, FIG. **4** and the following discussion are intended to provide a brief, general description of a suitable computing environment **400** in which the various embodiments of the subject disclosure can be implemented. In particular, computing environment **400** can be used in the implementation of network elements **150**, **152**, **154**, **156**, access terminal **112**, base station or access point **122**, switching device **132**, media terminal **142**, and/or VNEs **330**, **332**, **334**, etc. Each of these devices can be implemented via computer-executable instructions that can run on one or more computers, and/or in combination with other

program modules and/or as a combination of hardware and software. For example, computing environment **400** can facilitate in whole or in part a LEO satellite emergency communications system and method.

[0087] Generally, program modules comprise routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0088] As used herein, a processing circuit includes one or more processors as well as other application specific circuits such as an application specific integrated circuit, digital logic circuit, state machine, programmable gate array or other circuit that processes input signals or data and that produces output signals or data in response thereto. It should be noted that while any functions and features described herein in association with the operation of a processor could likewise be performed by a processing circuit.

[0089] The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0090] Computing devices typically comprise a variety of media, which can comprise computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and comprises both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data or unstructured data.

[0091] Computer-readable storage media can comprise, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0092] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0093] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and comprises any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media comprise wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

[0094] With reference again to FIG. **4**, the example environment can comprise a computer **402**, the

computer **402** comprising a processing unit **404**, a system memory **406** and a system bus **408**. The system bus **408** couples system components including, but not limited to, the system memory **406** to the processing unit **404**. The processing unit **404** can be any of various commercially available processors. Dual microprocessors and other multiprocessor architectures can also be employed as the processing unit **404**.

[0095] The system bus **408** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory **406** comprises ROM **410** and RAM **412**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **402**, such as during startup. The RAM **412** can also comprise a high-speed RAM such as static RAM for caching data.

[0096] The computer **402** further comprises an internal hard disk drive (HDD) **414** (e.g., EIDE, SATA), which internal HDD **414** can also be configured for external use in a suitable chassis (not shown), a magnetic floppy disk drive (FDD) **416**, (e.g., to read from or write to a removable diskette **418**) and an optical disk drive **420**, (e.g., reading a CD-ROM disk **422** or, to read from or write to other high-capacity optical media such as the DVD). The HDD **414**, magnetic FDD **416** and optical disk drive **420** can be connected to the system bus **408** by a hard disk drive interface **424**, a magnetic disk drive interface **426** and an optical drive interface **428**, respectively. The hard disk drive interface **424** for external drive implementations comprises at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0097] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **402**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to a hard disk drive (HDD), a removable magnetic diskette, and a removable optical media such as a CD or DVD, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, such as zip drives, magnetic cassettes, flash memory cards, cartridges, and the like, can also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0098] A number of program modules can be stored in the drives and RAM **412**, comprising an operating system **430**, one or more application programs **432**, other program modules **434** and program data **436**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **412**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0099] A user can enter commands and information into the computer **402** through one or more wired/wireless input devices, e.g., a keyboard **438** and a pointing device, such as a mouse **440**. Other input devices (not shown) can comprise a microphone, an infrared (IR) remote control, a joystick, a game pad, a stylus pen, touch screen or the like. These and other input devices are often connected to the processing unit **404** through an input device interface **442** that can be coupled to the system bus **408**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a universal serial bus (USB) port, an IR interface, etc.

[0100] A monitor **444** or other type of display device can be also connected to the system bus **408** via an interface, such as a video adapter **446**. It will also be appreciated that in alternative embodiments, a monitor **444** can also be any display device (e.g., another computer having a display, a smart phone, a tablet computer, etc.) for receiving display information associated with computer **402** via any communication means, including via the Internet and cloud-based networks.

In addition to the monitor **444**, a computer typically comprises other peripheral output devices (not shown), such as speakers, printers, etc.

[0101] The computer **402** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **448**. The remote computer(s) **448** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically comprises many or all of the elements described relative to the computer **402**, although, for purposes of brevity, only a remote memory/storage device **450** is illustrated. The logical connections depicted comprise wired/wireless connectivity to a local area network (LAN) **452** and/or larger networks, e.g., a wide area network (WAN) **454**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

[0102] When used in a LAN networking environment, the computer **402** can be connected to the LAN **452** through a wired and/or wireless communication network interface or adapter **456**. The adapter **456** can facilitate wired or wireless communication to the LAN **452**, which can also comprise a wireless AP disposed thereon for communicating with the adapter **456**.

[0103] When used in a WAN networking environment, the computer **402** can comprise a modem **458** or can be connected to a communications server on the WAN **454** or has other means for establishing communications over the WAN **454**, such as by way of the Internet. The modem **458**, which can be internal or external and a wired or wireless device, can be connected to the system bus **408** via the input device interface **442**. In a networked environment, program modules depicted relative to the computer **402** or portions thereof, can be stored in the remote memory/storage device **450**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

[0104] The computer **402** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This can comprise Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0105] Wi-Fi can allow connection to the Internet from a couch at home, a bed in a hotel room or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE 802.11 (a, b, g, n, ac, ag, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which can use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands for example or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 10BaseT wired Ethernet networks used in many offices.

[0106] Turning now to FIG. 5, an embodiment **500** of a mobile network platform **510** is shown that is an example of network elements **150**, **152**, **154**, **156**, and/or VNEs **330**, **332**, **334**, etc. For example, platform **510** can facilitate in whole or in part a LEO satellite emergency communications system and method. In one or more embodiments, the mobile network platform **510** can generate and receive signals transmitted and received by base stations or access points such as base station or access point **122**. Generally, mobile network platform **510** can comprise components, e.g., nodes, gateways, interfaces, servers, or disparate platforms, that facilitate both packet-switched (PS) (e.g., internet protocol (IP), frame relay, asynchronous transfer mode (ATM)) and circuit-

switched (CS) traffic (e.g., voice and data), as well as control generation for networked wireless telecommunication. As a non-limiting example, mobile network platform **510** can be included in telecommunications carrier networks and can be considered carrier-side components as discussed elsewhere herein. Mobile network platform **510** comprises CS gateway node(s) **512** which can interface CS traffic received from legacy networks like telephony network(s) **540** (e.g., public switched telephone network (PSTN), or public land mobile network (PLMN)) or a signaling system #7 (SS7) network **560**. CS gateway node(s) **512** can authorize and authenticate traffic (e.g., voice) arising from such networks. Additionally, CS gateway node(s) **512** can access mobility, or roaming, data generated through SS7 network **560**; for instance, mobility data stored in a visited location register (VLR), which can reside in memory **530**. Moreover, CS gateway node(s) **512** interfaces CS-based traffic and signaling and PS gateway node(s) **518**. As an example, in a 3GPP UMTS network, CS gateway node(s) **512** can be realized at least in part in gateway GPRS support node(s) (GGSN). It should be appreciated that functionality and specific operation of CS gateway node(s) **512**, PS gateway node(s) **518**, and serving node(s) **516**, is provided and dictated by radio technology(ies) utilized by mobile network platform **510** for telecommunication over a radio access network **520** with other devices, such as a radiotelephone **575**.

[0107] In addition to receiving and processing CS-switched traffic and signaling, PS gateway node(s) **518** can authorize and authenticate PS-based data sessions with served mobile devices. Data sessions can comprise traffic, or content(s), exchanged with networks external to the mobile network platform **510**, like wide area network(s) (WANs) **550**, enterprise network(s) **570**, and service network(s) **580**, which can be embodied in local area network(s) (LANs), can also be interfaced with mobile network platform **510** through PS gateway node(s) **518**. It is to be noted that WANs **550** and enterprise network(s) **570** can embody, at least in part, a service network(s) like IP multimedia subsystem (IMS). Based on radio technology layer(s) available in technology resource(s) or radio access network **520**, PS gateway node(s) **518** can generate packet data protocol contexts when a data session is established; other data structures that facilitate routing of packetized data also can be generated. To that end, in an aspect, PS gateway node(s) **518** can comprise a tunnel interface (e.g., tunnel termination gateway (TTG) in 3GPP UMTS network(s) (not shown)) which can facilitate packetized communication with disparate wireless network(s), such as Wi-Fi networks.

[0108] In embodiment **500**, mobile network platform **510** also comprises serving node(s) **516** that, based upon available radio technology layer(s) within technology resource(s) in the radio access network **520**, convey the various packetized flows of data streams received through PS gateway node(s) **518**. It is to be noted that for technology resource(s) that rely primarily on CS communication, server node(s) can deliver traffic without reliance on PS gateway node(s) **518**; for example, server node(s) can embody at least in part a mobile switching center. As an example, in a 3GPP UMTS network, serving node(s) **516** can be embodied in serving GPRS support node(s) (SGSN).

[0109] For radio technologies that exploit packetized communication, server(s) **514** in mobile network platform **510** can execute numerous applications that can generate multiple disparate packetized data streams or flows, and manage (e.g., schedule, queue, format . . .) such flows. Such application(s) can comprise add-on features to standard services (for example, provisioning, billing, customer support . . .) provided by mobile network platform **510**. Data streams (e.g., content(s) that are part of a voice call or data session) can be conveyed to PS gateway node(s) **518** for authorization/authentication and initiation of a data session, and to serving node(s) **516** for communication thereafter. In addition to application server, server(s) **514** can comprise utility server(s), a utility server can comprise a provisioning server, an operations and maintenance server, a security server that can implement at least in part a certificate authority and firewalls as well as other security mechanisms, and the like. In an aspect, security server(s) secure communication served through mobile network platform **510** to ensure network's operation and data integrity in

addition to authorization and authentication procedures that CS gateway node(s) **512** and PS gateway node(s) **518** can enact. Moreover, provisioning server(s) can provision services from external network(s) like networks operated by a disparate service provider; for instance, WAN **550** or Global Positioning System (GPS) network(s) (not shown). Provisioning server(s) can also provision coverage through networks associated to mobile network platform **510** (e.g., deployed and operated by the same service provider), such as the distributed antennas networks shown in FIG. **1(s)** that enhance wireless service coverage by providing more network coverage.

[0110] It is to be noted that server(s) **514** can comprise one or more processors configured to confer at least in part the functionality of mobile network platform **510**. To that end, the one or more processors can execute code instructions stored in memory **530**, for example. It should be appreciated that server(s) **514** can comprise a content manager, which operates in substantially the same manner as described hereinbefore.

[0111] In example embodiment **500**, memory **530** can store information related to operation of mobile network platform **510**. Other operational information can comprise provisioning information of mobile devices served through mobile network platform **510**, subscriber databases; application intelligence, pricing schemes, e.g., promotional rates, flat-rate programs, couponing campaigns; technical specification(s) consistent with telecommunication protocols for operation of disparate radio, or wireless, technology layers; and so forth. Memory **530** can also store information from at least one of telephony network(s) **540**, WAN **550**, SS7 network **560**, or enterprise network(s) **570**. In an aspect, memory **530** can be, for example, accessed as part of a data store component or as a remotely connected memory store.

[0112] In order to provide a context for the various aspects of the disclosed subject matter, FIG. **5**, and the following discussion, are intended to provide a brief, general description of a suitable environment in which the various aspects of the disclosed subject matter can be implemented. While the subject matter has been described above in the general context of computer-executable instructions of a computer program that runs on a computer and/or computers, those skilled in the art will recognize that the disclosed subject matter also can be implemented in combination with other program modules. Generally, program modules comprise routines, programs, components, data structures, etc. that perform particular tasks and/or implement particular abstract data types.

[0113] Turning now to FIG. **6**, an illustrative embodiment of a communication device **600** is shown. The communication device **600** can serve as an illustrative embodiment of devices such as data terminals **114**, mobile devices **124**, vehicle **126**, display devices **144** or other client devices for communication via either communications network **125**. For example, computing device **600** can facilitate in whole or in part a LEO satellite emergency communications system and method.

[0114] The communication device **600** can comprise a wireline and/or wireless transceiver **602** (herein transceiver **602**), a user interface (UI) **604**, a power supply **614**, a location receiver **616**, a motion sensor **618**, an orientation sensor **620**, and a controller **606** for managing operations thereof. The transceiver **602** can support short-range or long-range wireless access technologies such as Bluetooth®, ZigBee®, Wi-Fi, DECT, or cellular communication technologies, just to mention a few (Bluetooth® and ZigBee® are trademarks registered by the Bluetooth® Special Interest Group and the ZigBee® Alliance, respectively). Cellular technologies can include, for example, CDMA-1×, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, WiMAX, SDR, LTE, as well as other next generation wireless communication technologies as they arise. The transceiver **602** can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched wireline access technologies (such as TCP/IP, VOIP, etc.), and combinations thereof.

[0115] The UI **604** can include a depressible or touch-sensitive keypad **608** with a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device **600**. The keypad **608** can be an integral part of a housing assembly of the communication device **600** or an independent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example

Bluetooth®. The keypad **608** can represent a numeric keypad commonly used by phones, and/or a QWERTY keypad with alphanumeric keys. The UI **604** can further include a display **610** such as monochrome or color LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device **600**. In an embodiment where the display **610** is touch-sensitive, a portion or all of the keypad **608** can be presented by way of the display **610** with navigation features.

[0116] The display **610** can use touch screen technology to also serve as a user interface for detecting user input. As a touch screen display, the communication device **600** can be adapted to present a user interface having graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The display **610** can be equipped with capacitive, resistive or other forms of sensing technology to detect how much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used to control the manipulation of the GUI elements or other functions of the user interface. The display **610** can be an integral part of the housing assembly of the communication device **600** or an independent device communicatively coupled thereto by a tethered wireline interface (such as a cable) or a wireless interface.

[0117] The UI **604** can also include an audio system **612** that utilizes audio technology for conveying low volume audio (such as audio heard in proximity of a human ear) and high-volume audio (such as speakerphone for hands free operation). The audio system **612** can further include a microphone for receiving audible signals of an end user. The audio system **612** can also be used for voice recognition applications. The UI **604** can further include an image sensor **613** such as a charged coupled device (CCD) camera for capturing still or moving images.

[0118] The power supply **614** can utilize common power management technologies such as replaceable and rechargeable batteries, supply regulation technologies, and/or charging system technologies for supplying energy to the components of the communication device **600** to facilitate long-range or short-range portable communications. Alternatively, or in combination, the charging system can utilize external power sources such as DC power supplied over a physical interface such as a USB port or other suitable tethering technologies.

[0119] The location receiver **616** can utilize location technology such as a global positioning system (GPS) receiver capable of assisted GPS for identifying a location of the communication device **600** based on signals generated by a constellation of GPS satellites, which can be used for facilitating location services such as navigation. The motion sensor **618** can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing technology to detect motion of the communication device **600** in three-dimensional space. The orientation sensor **620** can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device **600** (north, south, west, and east, as well as combined orientations in degrees, minutes, or other suitable orientation metrics).

[0120] The communication device **600** can use the transceiver **602** to also determine a proximity to a cellular, Wi-Fi, Bluetooth®, or other wireless access points by sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or signal time of arrival (TOA) or time of flight (TOF) measurements. The controller **606** can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), programmable gate arrays, application specific integrated circuits, and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies for executing computer instructions, controlling, and processing data supplied by the aforementioned components of the communication device **600**.

[0121] Other components not shown in FIG. **6** can be used in one or more embodiments of the subject disclosure. For instance, the communication device **600** can include a slot for adding or removing an identity module such as a Subscriber Identity Module (SIM) card or Universal Integrated Circuit Card (UICC). SIM or UICC cards can be used for identifying subscriber

services, executing programs, storing subscriber data, and so on.

[0122] The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and does not otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

[0123] In the subject specification, terms such as “store,” “storage,” “data store,” data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can comprise both volatile and nonvolatile memory, by way of illustration, and not limitation, volatile memory, non-volatile memory, disk storage, and memory storage. Further, nonvolatile memory can be included in read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), or flash memory. Volatile memory can comprise random access memory (RAM), which acts as external cache memory. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SLDRAM), and direct Rambus RAM (DRRAM). Additionally, the disclosed memory components of systems or methods herein are intended to comprise, without being limited to comprising, these and any other suitable types of memory.

[0124] Moreover, it will be noted that the disclosed subject matter can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as personal computers, hand-held computing devices (e.g., PDA, phone, smartphone, watch, tablet computers, netbook computers, etc.), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distributed computing environments where tasks are performed by remote processing devices that are linked through a communications network; however, some if not all aspects of the subject disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0125] In one or more embodiments, information regarding use of services can be generated including services being accessed, media consumption history, user preferences, and so forth. This information can be obtained by various methods including user input, detecting types of communications (e.g., video content vs. audio content), analysis of content streams, sampling, and so forth. The generating, obtaining and/or monitoring of this information can be responsive to an authorization provided by the user. In one or more embodiments, an analysis of data can be subject to authorization from user(s) associated with the data, such as an opt-in, an opt-out, acknowledgement requirements, notifications, selective authorization based on types of data, and so forth.

[0126] Some of the embodiments described herein can also employ artificial intelligence (AI) to facilitate automating one or more features described herein. The embodiments (e.g., in connection with automatically facilitating a LEO satellite emergency communications system and method) can employ various AI-based schemes for carrying out various embodiments thereof. Moreover, the classifier can be employed to determine a ranking or priority of each satellite and/or end-user device. A classifier is a function that maps an input attribute vector, $x=(x_{\text{sub.1}}, x_{\text{sub.2}}, x_{\text{sub.3}}, x_{\text{sub.4}} \dots x_{\text{sub.n}})$, to a confidence that the input belongs to a class, that is, $f(x)=\text{confidence}(\text{class})$. Such classification can employ a probabilistic and/or statistical-based analysis (e.g., factoring into the analysis utilities and costs) to determine or infer an action that a user desires to be

automatically performed. A support vector machine (SVM) is an example of a classifier that can be employed. The SVM operates by finding a hypersurface in the space of possible inputs, which the hypersurface attempts to split the triggering criteria from the non-triggering events. Intuitively, this makes the classification correct for testing data that is near, but not identical to training data. Other directed and undirected model classification approaches comprise, e.g., naïve Bayes, Bayesian networks, decision trees, neural networks, fuzzy logic models, and probabilistic classification models providing different patterns of independence can be employed. Classification as used herein also is inclusive of statistical regression that is utilized to develop models of priority.

[0127] As will be readily appreciated, one or more of the embodiments can employ classifiers that are explicitly trained (e.g., via a generic training data) as well as implicitly trained (e.g., via observing UE behavior, operator preferences, historical information, receiving extrinsic information). For example, SVMs can be configured via a learning or training phase within a classifier constructor and feature selection module. Thus, the classifier(s) can be used to automatically learn and perform a number of functions, including but not limited to determining according to predetermined criteria which of the satellite(s) and/or end-user device(s) is to receive priority.

[0128] As used in some contexts in this application, in some embodiments, the terms “component,” “system” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

[0129] Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device or computer-readable storage/communications media. For example, computer readable storage media can include, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications

can be made to this configuration without departing from the scope or spirit of the various embodiments.

[0130] In addition, the words “example” and “exemplary” are used herein to mean serving as an instance or illustration. Any embodiment or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

[0131] Moreover, terms such as “user equipment,” “mobile station,” “mobile,” subscriber station,” “access terminal,” “terminal,” “handset,” “mobile device” (and/or terms representing similar terminology) can refer to a wireless device utilized by a subscriber or user of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings.

[0132] Furthermore, the terms “user,” “subscriber,” “customer,” “consumer” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based, at least, on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

[0133] As employed herein, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units.

[0134] As used herein, terms such as “data storage,” data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components or computer-readable storage media, described herein can be either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory.

[0135] What has been described above includes mere examples of various embodiments. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing these examples, but one of ordinary skill in the art can recognize that many further combinations and permutations of the present embodiments are possible. Accordingly, the embodiments disclosed and/or claimed herein are intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims.

Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

[0136] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

[0137] As may also be used herein, the term(s) “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via one or more intervening items. Such items and intervening items include, but are not limited to, junctions, communication paths, components, circuit elements, circuits, functional blocks, and/or devices. As an example of indirect coupling, a signal conveyed from a first item to a second item may be modified by one or more intervening items by modifying the form, nature or format of information in a signal, while one or more elements of the information in the signal are nevertheless conveyed in a manner than can be recognized by the second item. In a further example of indirect coupling, an action in a first item can cause a reaction on the second item, as a result of actions and/or reactions in one or more intervening items.

[0138] Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement which achieves the same or similar purpose may be substituted for the embodiments described or shown by the subject disclosure. The subject disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. For instance, one or more features from one or more embodiments can be combined with one or more features of one or more other embodiments. In one or more embodiments, features that are positively recited can also be negatively recited and excluded from the embodiment with or without replacement by another structural and/or functional feature. The steps or functions described with respect to the embodiments of the subject disclosure can be performed in any order. The steps or functions described with respect to the embodiments of the subject disclosure can be performed alone or in combination with other steps or functions of the subject disclosure, as well as from other embodiments or from other steps that have not been described in the subject disclosure. Further, more than or less than all of the features described with respect to an embodiment can also be utilized.

Claims

1. A method, comprising: determining whether a backhaul communication path between a wireless communications node and a core network has become degraded, wherein the wireless communications node comprises at least one radio element configured for wireless communications with a plurality of wireless end-user devices, and wherein the determining results in a first determination; responsive to the first determination being that the backhaul communication path has become degraded, connecting a Low Earth Orbit (LEO) satellite antenna for first bi-directional communication between the LEO satellite antenna and the at least one radio element; configuring the LEO satellite antenna for second bi-directional communication between the LEO satellite antenna and a LEO satellite, wherein the second bi-directional communication comprises a first uplink to the LEO satellite and a first downlink from the LEO satellite, wherein the LEO satellite is configured for third bi-directional communication with the core network, and

wherein the third bi-directional communication comprises a second downlink from the LEO satellite and a second uplink to the LEO satellite; and based upon the configuring, facilitating outgoing communications from each of the plurality of wireless end-user devices to the core network via the LEO satellite and facilitating incoming communications to each of the plurality of wireless end-user devices from the core network via the LEO satellite.

2. The method of claim 1, wherein the backhaul communication path comprises a fiber optic link, a microwave link, or any combination thereof.
3. The method of claim 1, wherein the determining whether the backhaul communication path has become degraded comprises determining whether the backhaul communication path has degraded below a threshold.
4. The method of claim 1, wherein the determining whether the backhaul communication path has become degraded comprises determining whether the backhaul communication path has become entirely unusable for carrying communications.
5. The method of claim 1, wherein the wireless communications node comprises a cellular base station.
6. The method of claim 5, wherein: the cellular base station comprises a tower; and the method further comprises physically mounting the LEO satellite antenna to the tower.
7. The method of claim 1, wherein: the wireless communications node further comprises at least one router and at least one baseband unit; the first bi-directional communication between the LEO satellite antenna and the at least one radio element is carried out via the at least one router and the at least one baseband unit; the at least one router is connected to the at least one baseband unit by one or more first electrical wires, one or more first waveguides, one or more first fiber optic cables, or any combination thereof; and the at least one baseband unit is connected to the at least one radio element by one or more second electrical wires, one or more second waveguides, one or more second fiber optic cables, or any combination thereof.
8. The method of claim 1, wherein the configuring the LEO satellite antenna further comprises configuring the LEO satellite antenna for the second bi-directional communication between the LEO satellite antenna and a plurality of LEO satellites including the LEO satellite.
9. The method of claim 8, wherein the configuring the LEO satellite antenna further comprises configuring the LEO satellite antenna for the second bi-directional communication between the LEO satellite antenna and each of the plurality of LEO satellites in succession.
10. The method of claim 1, further comprising: further determining whether the backhaul communication path between the wireless communications node and the core network is no longer degraded, wherein the further determining results in a second determination; responsive to the second determination being that the backhaul communication path is no longer degraded, disconnecting the LEO satellite antenna from the first bi-directional communication between the LEO satellite antenna and the at least one radio element; and facilitating subsequent outgoing communications from each of the plurality of wireless end-user devices to the core network via the backhaul communication path and facilitating subsequent incoming communications to each of the plurality of wireless end-user devices from the core network via the backhaul communication path.
11. The method of claim 10, wherein the further determining whether the backhaul communication path between the wireless communications node and the core network is no longer degraded comprises further determining whether the backhaul communication path is no longer degraded below a threshold.
12. The method of claim 1, wherein: the second downlink is from the LEO satellite to a LEO ground station; and the second uplink is from the LEO ground station to the LEO satellite.
13. The method of claim 12, wherein the LEO ground station communicates with the core network via the Internet.
14. The method of claim 13, wherein the LEO ground station communicates with the core network through a secure gateway.

15. The method of claim 1, wherein the core network is part of a cellular carrier network.

16. The method of claim 1, wherein each of the plurality of wireless end-user devices comprises a respective smartphone, a respective cell phone, a respective tablet computer, a respective laptop computer, or a respective combination thereof.

17. A method, comprising: determining whether a backhaul communication path between a cell site and a core service provider network has become degraded, wherein the cell site comprises at least one cellular radio configured for wireless communications with at least one end-user mobile communication device, wherein the cell site further comprises at least one baseband unit and at least one router, and wherein the determining results in a first determination; responsive to the first determination being that the backhaul communication path has become degraded, installing a Low Earth Orbit (LEO) satellite antenna at the cell site for first bi-directional communication between the LEO satellite antenna and the at least one cellular radio via the at least one router and the at least one baseband unit; configuring the LEO satellite antenna for second bi-directional communication between the LEO satellite antenna and a LEO satellite, wherein the second bi-directional communication comprises a first uplink to the LEO satellite and a first downlink from the LEO satellite, wherein the LEO satellite is configured for third bi-directional communication with the core service provider network, and wherein the third bi-directional communication comprises a second downlink from the LEO satellite and a second uplink to the LEO satellite; and based upon the configuring, facilitating an outgoing communication from the at least one end-user mobile communication device to the core service provider network via the LEO satellite and facilitating an incoming communication to the at least one end-user mobile communication device from the core service provider network via the LEO satellite.

18. The method of claim 17, wherein: the cell site has a source of electrical power; the LEO antenna has one or more electrical components associated therewith; and the installing the LEO satellite antenna at the cell site comprises connecting the one or more electrical components to the source of electrical power.

19. A method, comprising: responsive to determining that a backhaul communication path between a cellular base station and a core network has become degraded, installing a Low Earth Orbit (LEO) satellite antenna on a tower of the cellular base station for first bi-directional communication between the LEO satellite antenna and a cellular radio of the cellular base station, wherein the first bi-directional communication is via a router of the cellular base station and a baseband unit of the cellular base station, and wherein the cellular radio is configured for wireless communications with a plurality of end-user cellular communication devices; configuring the LEO satellite antenna for second bi-directional communication between the LEO satellite antenna and a LEO satellite, wherein the second bi-directional communication comprises a first uplink to the LEO satellite and a first downlink from the LEO satellite, wherein the LEO satellite is configured for third bi-directional communication with the core network, and wherein the third bi-directional communication comprises a second downlink from the LEO satellite and a second uplink to the LEO satellite; and based upon the configuring, implementing outgoing communications from the plurality of end-user cellular communication devices to the core network via the LEO satellite and implementing incoming communications to the plurality of end-user cellular communication devices from the core network via the LEO satellite.

20. The method of claim 19, wherein: the backhaul communication path comprises a fiber optic link, a microwave link, or any combination thereof; and the method further comprises: responsive to further determining that the backhaul communication path between the cellular base station and the core network is no longer degraded, disconnecting the LEO satellite antenna from the first bi-directional communication; and implementing subsequent outgoing communications from the plurality of end-user cellular communication devices to the core network via the backhaul communication path instead of via the LEO satellite and implementing subsequent incoming

communications to the plurality of end-user cellular communication devices from the core network via the backhaul communication path instead of via the LEO satellite.
